

GT1151Q

Programming Guide

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Address: Floor 13, Phase B, Tengfei Industrial Building, Futian Free Trade Zone, Shenzhen, China.
Tel: +86-755-33338828

Fax: +86-755-33338788

Post code: 518000

Email: info@goodix.com

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1. I²C Interface

GT1151Q interfaces with the host via 6 pins: VDD, GND, SCL, SDA, INT and RESET.

The INT (Interrupt) pin of the host can be falling-edge triggered. In addition, INT must be set as input and pulled up through the internal setting by host or external pull-up resistor. The host controls the RESET pin of the GT1151Q by driving it high or low. To ensure reliable reset, it is recommended that RESET pin be driven low for longer than 100μs.

GT1151Q communicates with host via standard I²C interface, with a maximum transmission rate of 400K bps. When the host communicates at rates exceeding 200K bps, it is required to pay special attention to the resistance of the external pull-up resistor of I²C interface to ensure the Rise time and Fall time of SCL and SDA signals comply with the requirements specified in GT1151Q datasheet. GT1151Q invariably serves as slave device in communication and its I²C slave address consists of 7 address bits and 1 Read/Write control bit. The high-order 7 bits are slave address while bit 0 is Read/Write control bit. The slave address of GT1151Q is shown below:

7-bit address	8-bit write address	8-bit read address
0x14	0x28	0x29

2. I²C Timings

2.1 Timing for Write Operation



S: Start condition.

Address_W: slave address with Write control bit.

ACK: Acknowledgement signal.

Register_H, Register_L: 16-bit register address from which Write Operation starts

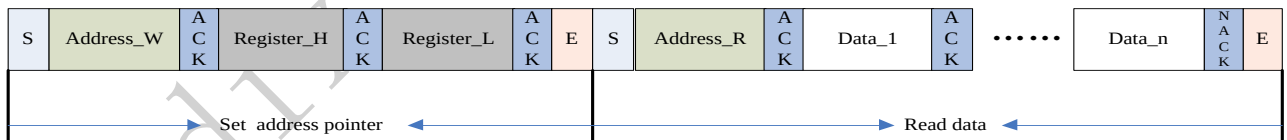
Data_1 to Data_n: Data bytes 1 to n.

E: Stop condition.

After setting the address pointer for Write operation, it is allowed to write one or more than one byte data at a time. GT1151Q will automatically update the address pointer and store the data bytes in sequence.

2.2 Timing for Read Operation

First, set address pointer based on the aforesaid Write Operation timing sequence. Then, resend Start condition to perform Read addressing and read data.



Address_R: Slave address with Read control bit.

NACK: Host issues NACK after reading the last byte.

After setting Read addresses, the host can read one or more than one byte at a time. GT1151Q will automatically update the address pointer and send subsequent data in sequence.

The Stop condition (the first E signal as shown in the above diagram) after setting the address pointer is optional. However, the repeated Start condition has to be sent.

3. Register Map

3.1 Real-time Command Registers (Write only)

Addr	Name	bit7	bit6	bit5	bit4	bit3	bit2	bit1	bit0
0x8040	Command	0x00: Read coordinates status; 0x01: Read diff data & raw data; 0x02: Read diff data & raw data; 0x03: baseline update (Internal test); 0x04: baseline calibration (Internal test); 0x05: Screen off; 0x06: Enter Charge mode; 0x07: Exit Charge mode ; 0x08 :Enter Gesture mode ; 0x0b: Finger touch detection mode (weak signal touch is not supported) 0x0c: Auto-switch mode (automatically switch from finger touch detection mode to gloved hand touch detection mode) 0x31: Save customized gesture templates; 0x35: Clear gesture templates in the touch controller; 0x36: Delete a certain gesture template; 0x37: Query the gesture template ; 0xAA: used by ESD protection mechanism; driver reads and checks the value of 0x8040, and writes 0xAA to 0x8040 periodically.							
0x8041	Command_Data	Data corresponds to command (for commands without data, please issue 0)							
0x8042	Command_Checksum	Checksum of the command and data [sum (0x8040~0x8042)==0]							
0x8043	ESD_Check	Used by ESD used by ESD protection mechanism; reset to 0 upon initialization; after that, the driver implements Read/Write operation.							
0x8044	Request	Request initiaively sent by the firmware to the host							
0x8045	FW_Status_L	Firmware status bytes							
0x8046	FW_Status_H								

3.2 Configuration Registers (R/W)

Register	Config Data	bit7	bit6	bit5	bit4	bit3	bit2	bit1	bit0
0x8050	Config_Version	Bit 7 is the stationary configuration flag (0: unfixed; 1: fixed); bit 0 to bit 6 indicate the version number of configuration document.							
0x8051 ~ 0x813B		Configuration content							
0x813C	Config_Chksum_H	Configuration 16-bit Checksum(big-endian mode: most significant byte is stored at the lowest memory address)							
0x813D	Config_Chksum_L								
0x813E	Config_Fresh	Configuration refreshed flag (the host writes 1 to this field)							

Additional configuration

Addr	Config Data	bit7	bit6	bit5	bit4	bit3	bit2	bit1	bit0
0xBF7B ~ 0xBFF8		Configuration content							
0xBFF9	Config_Chksum_H	Configuration 16-bit Checksum(big-endian mode: most significant byte is stored at the lowest memory address)							
0xBFFA	Config_Chksum_L								

3.3 Coordinate Registers

Addr	Access	bit7	bit6	bit5	bit4	bit3	bit2	bit1	bit0
0x8140	R	Product ID (first byte, ASCII)							
0x8141	R	Product ID (second byte, ASCII)							
0x8142	R	Product ID (third byte, ASCII)							
0x8143	R	Product ID (fourth byte, ASCII)							
0x8144	R	CID in Patch version number. CID is the software version identifier. It is 2 decimal digits (shown in BCD code when stored in software), ranging from 00 to 26. "0" indicates standard version and will not be displayed; 1 to 26 indicates customized version and will be converted into "A to Z" when displayed							
0x8145	R	Patch minor version number(2-digit compressed BCD code)							
0x8146	R	Patch sub-version number(2-digit compressed BCD code)							
0x8147	R	MASK major version number							

0x8148	R	MASK minor version number				
0x8149	R	MASK internal version number				
0x814A	R	BondingOption			SensorID	
0x814B	R	checksum				
0x814C	R	Reserved				
0x814D	R	Reserved				
0x814E	R/W	Buffer Status	Large Detect	Rsvd	Have Key	Number of Touch Points
0x814F	R	Touch Sta	Reserved			track_id
0x8150	R	point 1 x coordinate (low byte)				
0x8151	R	point 1 x coordinate (high byte)				
0x8152	R	point 1 y coordinate (low byte)				
0x8153	R	point 1 y coordinate (high byte)				
0x8154	R	Point 1 size (W)				
0x8155	R	point 1 size (H)				
0x8156	R	Reserved				
0x8157	R	Touch Sta	Reserved			track_id
0x8158	R	point 2 x coordinate (low byte)				
0x8159	R	point 2 x coordinate (high byte)				
0x815A	R	point 2 y coordinate (low byte)				
0x815B	R	point 2 y coordinate (high byte)				
0x815C	R	point 2 size (W)				
0x815D	R	point 2 size (H)				
0x815E	R	Reserved				
0x815F	R	Touch Sta	Reserved			track_id
0x8160	R	point 3 x coordinate (low byte)				
0x8161	R	point 3 x coordinate (high byte)				
0x8162	R	point 3 y coordinate (low byte)				
0x8163	R	point 3 y coordinate (high byte)				
0x8164	R	point 3 size (W)				
0x8165	R	point 3 size (H)				
0x8166	R	Reserved				
0x8167	R	Touch Sta	Reserved			track_id
0x8168	R	point 4 x coordinate (low byte)				
0x8169	R	point 4 x coordinate (high byte)				
0x816A	R	point 4 y coordinate (low byte)				
0x816B	R	point 4 y coordinate (high byte)				

0x816C	R	point 4 size (W)		
0x816D	R	point 4 size (H)		
0x816E	R	Reserved		
0x816F	R	Touch Sta	Reserved	track_id
0x8170	R	point 5 x coordinate (low byte)		
0x8171	R	point 5 x coordinate (high byte)		
0x8172	R	point 5 y coordinate (low byte)		
0x8173	R	point 5 y coordinate (high byte)		
0x8174	R	point 5 size (W)		
0x8175	R	point 5 size (H)		
0x8176	R	Reserved		
0x8177	R	Touch Sta	Reserved	track_id
0x8178	R	point 6 x coordinate (low byte)		
0x8179	R	point 6 x coordinate (high byte)		
0x817A	R	point 6 y coordinate (low byte)		
0x817B	R	point 6 y coordinate (high byte)		
0x817C	R	point 6 size (W)		
0x817D	R	point 6 size (H)		
0x817E	R	Reserved		
0x817F	R	Touch Sta	Reserved	track_id
0x8180	R	point 7 x coordinate (low byte)		
0x8181	R	point 7 x coordinate (high byte)		
0x8182	R	point 7 y coordinate (low byte)		
0x8183	R	point 7 y coordinate (high byte)		
0x8184	R	point 7 size (W)		
0x8185	R	point 7 size (H)		
0x8186	R	Reserved		
0x8187	R	Touch Sta	Reserved	track_id
0x8188	R	point 8 x coordinate (low byte)		
0x8189	R	point 8 x coordinate (high byte)		
0x818A	R	point 8 y coordinate (low byte)		
0x818B	R	point 8 y coordinate (high byte)		
0x818C	R	point 8 size (W)		
0x818D	R	point 8 size (H)		
0x818E	R	Reserved		
0x818F	R	Touch Sta	Reserved	track_id

0x8190	R	point 9 x coordinate (low byte)		
0x8191	R	point 9 x coordinate (high byte)		
0x8192	R	point 9 y coordinate (low byte)		
0x8193	R	point 9 y coordinate (high byte)		
0x8194	R	point 9 size (W)		
0x8195	R	point 9 size (H)		
0x8196	R	Reserved		
0x8197	R	Touch Sta	Reserved	track_id
0x8198	R	point 10 x coordinate (low byte)		
0x8199	R	point 10 x coordinate (high byte)		
0x819A	R	point 10 y coordinate (low byte)		
0x819B	R	point 10 y coordinate (high byte)		
0x819C	R	point 10 size (W)		
0x819D	R	point 10 size (H)		
0x819E	R	Reserved		
0x819F	R	KeyValue		
0x81A0	R	Checksum(sum(0x814E:cur,len)==0), length="Touch Points"*8+3		

Supplementary description on some registers:**[0x814A]****Bit3-Bit0: Vendor_ID**

The ID of the current module is codetermined by pins *sensor_opt1* and *sensor_opt2* in the circuit.

Different combinations of their connections can identify 6 sensor IDs as shown below:

sensor_opt1	sensor_opt2	Vendor_ID
GND	GND	0
VDDIO	GND	1
NC	GND	2
GND	300K	3
VDDIO	300K	4
NC	300K	5

[0x814E]

Bit7: Buffer status, 1 = coordinate (or key) is ready for host to read; 0 = coordinate (or key) is not ready and data is not valid. After reading coordinates, host should configure this flag (or the entire byte) to 0 via I²C.

Bit4: HaveKey, 1 : key is touched; 0 : key is released.

Bit3~0: Number of touch points.

[0x814F]

Bit7: touch_sta. 1 indicates high-sensitivity touch coordinate;
0 indicates normal-sensitivity touch coordinate.

Bit3~0: track id. The ID number of the touch point.

[0x8177]: KeyValue

The address of KeyValue is not fixed. Instead, it stays behind valid coordinates. For example, 0x8177 is the address of the KeyValue when there are 5 coordinates on TP. However, if there are 4 coordinates on TP, the address of the KeyValue will be 0x816F.

3.4 Gesture Registers**(Gesture Feature Registers: share the addresses with the Coordinate Registers)**

Addr	Access	bit7	bit6	bit5	bit4	bit3	bit2	bit1	bit0
0x8140	R	Gesture ID (first Byte, ASCII G)							
0x8141	R	Gesture ID (second Byte, ASCII E)							
0x8142	R	Gesture ID (third Byte, ASCII S)							
0x8143	R	Gesture ID (fourth Byte, ASCII T)							
0x8144	R	Gesture major version number							
0x8145	R	Gesture minor version number							
0x8146	R	Gesture internal version number							
0x8147	R	MASK major version number							
0x8148	R	MASK minor version number							
0x8149	R	MASK internal version number							
0x814A	R	BondingOption				Vendor_ID			
0x814B	R	checksum							
0x814C	R/W	Gesture types(ASCII character indicates 0x21 - 0x7F), swipe right(0xAA), swipe left(0xBB), swipe down(0xAB), swipe up(0xBA), double-tap(0xCC), single tap on touch key (0xC1, 0xC2, 0xC4, 0xC8, the low-order four bits store the KeyValue), customized gestures (0x01~0x0A)							
0x814D	R	The number of touch points of a gesture((coordinates stored at0xBDA8))							
0x814E	R	Buffer 1 (starting address: 0x8150); Protocol type: 0x01: single-stroke gesture; 0x02: multi-stroke gesture.				Buffer 2 (starting address: 0xBDA8); Protocol type: 0x02: report all points; 0x02: report feature points.			
0x814F	R	The number of data bytes in Buffer 1							
0x8150 to 0x819E	R	Data in Buffer 1							
0x8151	R	Check Sum (Address location varies according to buffer length) Check start address: 0x814C Check length (u8): Gesture Data length(Buffer length) + 4							
0x819F	R	16 bits Check Sum of gesture trajectory coordinates (low byte) (Fixed address) Check start address: 0xBDA8 Check length (u16):Gesture Points Count * 2							
0x81A0	R	16 bits Check Sum of gesture trajectory coordinates (high byte) (Fixed address)							

(Gesture Coordinate Registers)

Addr	Access	bit7	bit6	bit5	bit4	bit3	bit2	bit1	bit0
0xBDA8	R	Gesture point 1 x coordinate(low byte)							
0xBDA9	R	Gesture point 1 x coordinate(high byte)							
0xBDAA	R	Gesture point 1 y coordinate(low byte)							
0BDAB	R	Gesture point 1 y coordinate(high byte)							

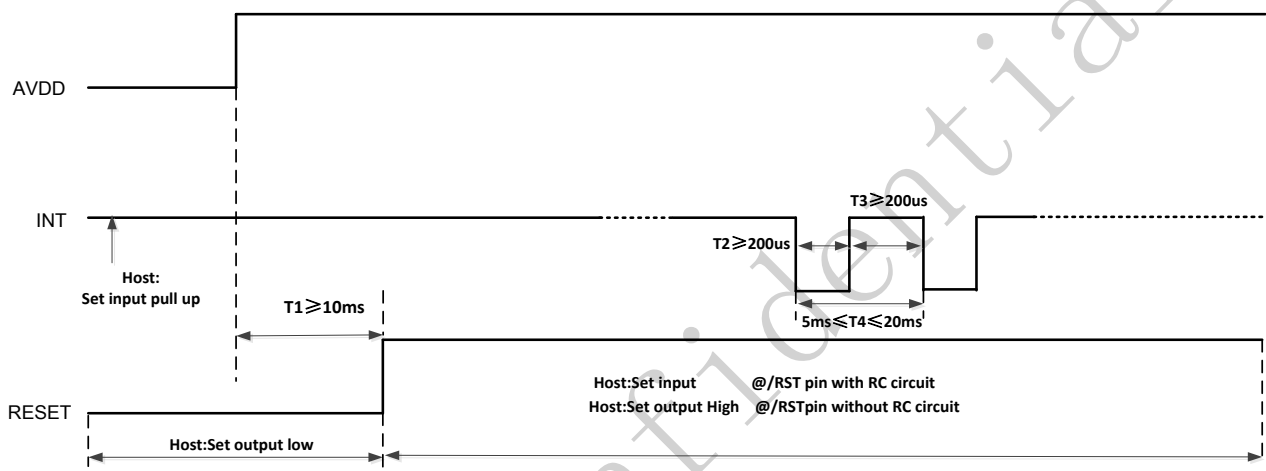
0xBDAC to 0xBEA7	R	Gesture point 2~64 coordinate(the number of coordinates equals to the value of 0x814D)
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4. Related Operation Timing

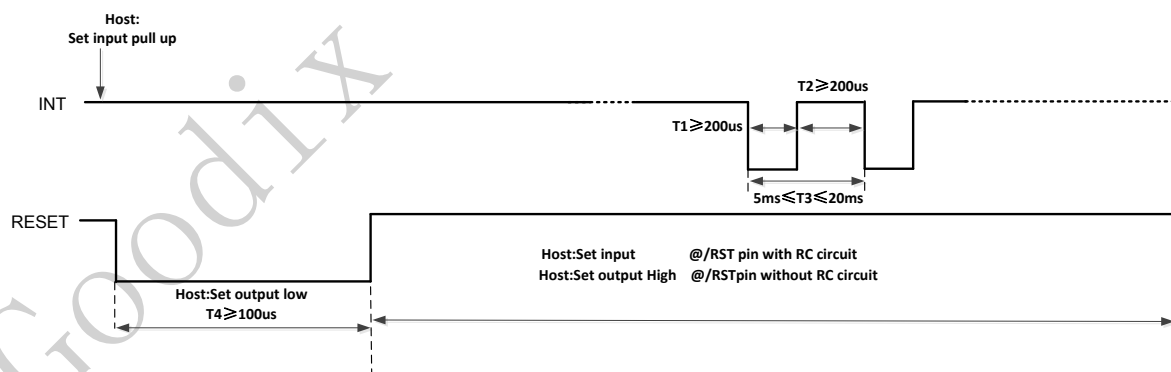
4.1 Power-On Timing

After power-on, the host needs to control such GT1151Q pins as AVDD, INT and RESET according to the timing sequence shown below:



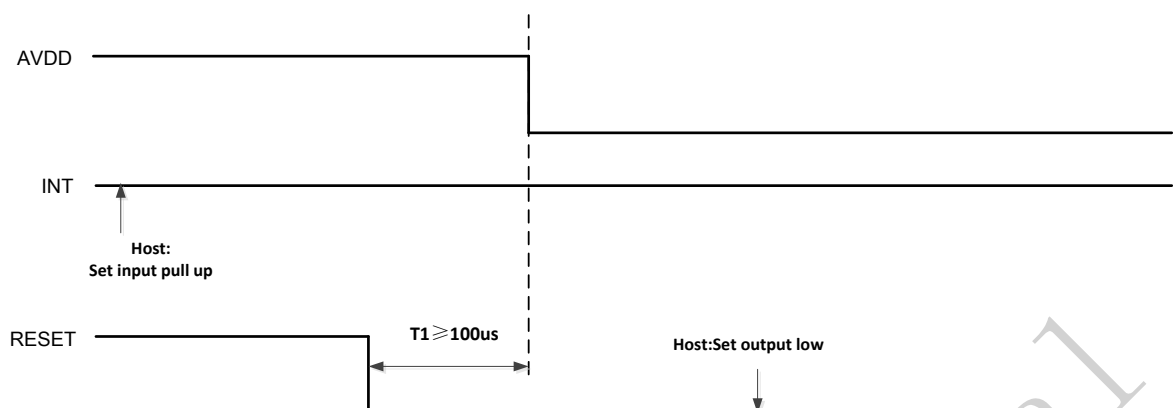
4.2 Reset Timing

Reset Timing diagram (the host resets GT1151Q):



4.3 Power-Down Timing

GT1151Q Power-Down Timing diagram:



4.4 Send Configuration after Power-on

During the power-on process, after host pulls RESET high, it is required to wait for 60ms before deciding whether to send the configuration. If request “0x01” is received, send the configuration; otherwise, it is unnecessary to do so.

4.5 Host responds to the “INT Request”

1) Description on the Request Data

Data	Description
0x00	The ACK sent by host to touch IC
0x01	Request host to send configuration
0x03	Request host to reset GT1151Q
0xFF	Idle
Others	Reserved

2) Respond to Request “0x01”

- If the interrupt is received and $0x814E=0$, the host reads $0x8044$ (“Request” register).
“ $0x8044=0x01$ ” indicates the request “0x01”.
- Write the configuration parameters to the configuration area via I^2C (please refer to section 3.2).
- Write a “0” to $0x8044$ via I^2C . The implementation of request “0x01” is finished.

3) Respond to Request “0x03”

- a) If the interrupt is received and 0x814E=0, the host reads 0x8044 ("Request" register).
"0x8044=0x03" indicates the request "0x03".
- b) Reset GT1151Q according to the Reset timing sequence. The implementation of request "0x03" is finished.

4.6 Register Value Modification

GT1151Q supports Register Value Modification. When modifying any register in the configuration area (Including regular and extended configurations) based on the timing sequence as specified in section 2, it is required to update the corresponding Config_Chksum and set Config_Fresh (0x813E) to 1 at the end. Otherwise, the modification is invalid; when modifying any registers outside configuration area, it is unnecessary to modify Config_Chksum and Config_Fresh.

5. Coordinates Reading

GT1151Q supports reading coordinate by periodic polling and interrupt request. To lighten the burden on the host, INT request is recommended.

When periodic polling is employed, the host reads coordinates through the following steps:

Based on the timing sequence specified in section 2, the host first reads register 0x814E.

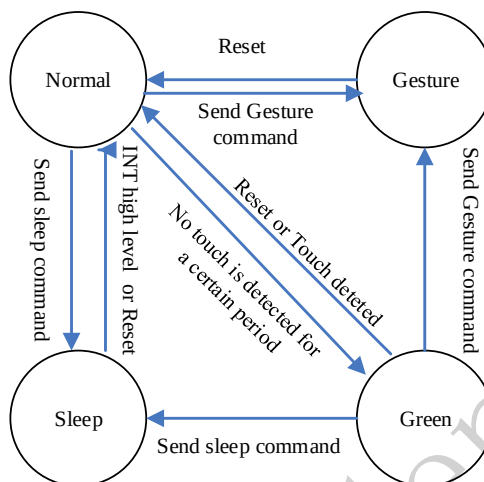
- a) If the data in buffer is ready (buffer status= 1), it reads coordinate and touch key information based on finger touch number and touch key status.
- b) If it is found out in step 1 that data in buffer is not ready (buffer status= 0), it will read again 1ms later.

The timing sequence for triggering an interrupt:

- 1) In standby mode, INT outputs high.
- 2) Drive INT from low to high (falling edge) when any coordinate is refreshed.
- 3) After the falling edge in step 2, INT will keep outputting low until next period (the period is configurable by setting Refresh_Rate). The host is supposed to finish reading the coordinate as soon as possible and reset buffer status (0x814E) to 0.
 - a) If the host fails to finish reading the coordinates within one scan cycle, GT1151Q will continue to perform coordinate calculation in next cycle and output falling edge instead of update the contents in the coordinate information area.
 - b) GT1151Q will not stop outputting INT pulse until the host reads the coordinate successfully.

6. Operation Modes Transition

6.1 Operating Modes



a) Normal Mode

When GT1151Q is operating in Normal mode, its fastest coordinate refresh period is 5ms-20ms (subject to configuration. One step is 1ms).

b) Green mode

When no touch is detected for a certain period (0s~14s, subject to configuration; one step is 1s) in Normal mode, GT1151Q will enter Green mode to reduce power consumption. Green Mode can be disabled through configuration.

In Green mode, the scan period for GT1151Q is about 40ms. It automatically enters Normal mode if any touch is detected.

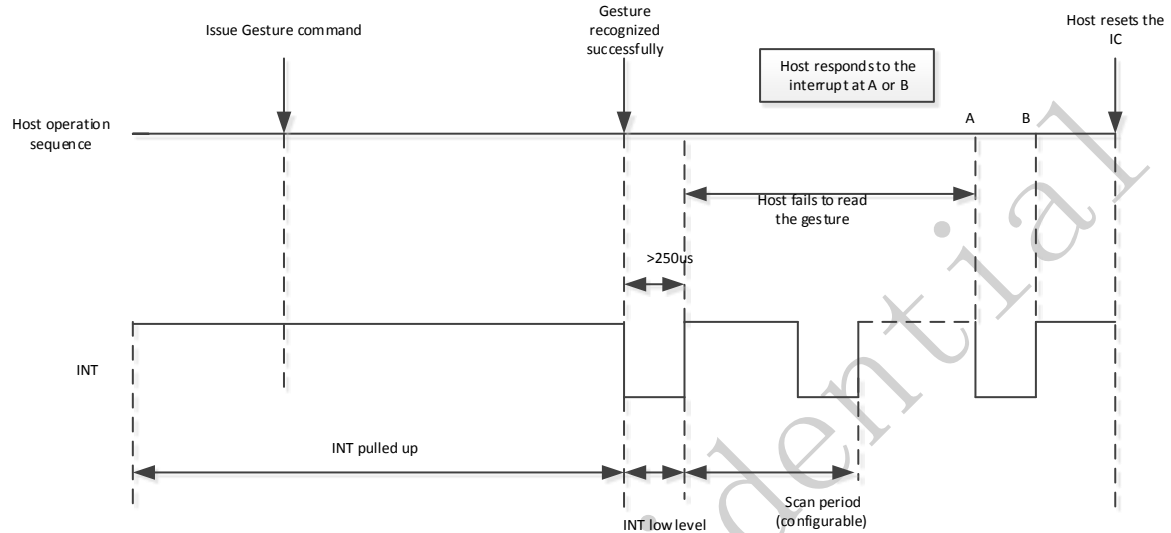
c) Gesture mode

After the host enables GT1151Q to enter Gesture mode by sending corresponding I²C command, wake-up can be achieved by swipe, double-tap, or writing of specified letters on the touch panel (TP).

In Gesture mode, when GT1151Q detects any finger swipe for a sufficiently long distance, double-tap, writing of specified letters on the TP, INT will output a low level pulse for longer than 250us (subject to configuration). The host turns on the screen after receiving such pulse and reset the GT1151Q to exit

Gesture Mode.

The timing when waking up the host by driving INT high:



d) Sleep mode

GT1151Q enters Sleep mode if it receives corresponding I²C command from the host

GT1151Q exits Sleep mode the host will reset GT1151Q. After the reset, GT1151Q will enter Normal mode. The time interval between issuing command and reset should be longer than 58ms.

7. Host System Driver Modification in Gesture Mode

7.1 Enter Gesture mode after Screen-off

- 1) If screen-off is achieved by pressing Power key (or any other key), send Command 0x08, 0x00 and 0XF8 to 0x8040, 0x8041 and 0x8042 respectively;
- 2) If screen-off is achieved due to timeout, send Command 0x08, 0x00 and 0XF8 to 0x8040, 0x8041 and 0x8042 respectively;
- 3) When the screen is off, if there is swipe, double-tap or writing of specified letters on the touch panel, INT pin will output a pulse for longer than 250us (subject to configuration) to notify the host. The host reads the value of 0x814C after receiving such pulse. If the value meets wake-up conditions, the host wakes up and turns on the screen. Otherwise, the host resets 0x814C and waits for the next pulse.

7.2 Enter Sleep Mode after Screen-off

- 1) If screen-off is achieved by pressing Power key (or any other key), send Command 0x05, 0x00 and 0XFB to 0x8040, 0x8041 and 0x8042, respectively;
- 2) If screen-off is achieved due to timeout, send Command 0x05, 0x00 and 0XFB to 0x8040, 0x8041 and 0x8042 respectively;
- 3) In Sleep mode, the host can be awakened only by pressing power key (or home key).

7.3 Wake up the host by pressing power key (or home key)

If awakened by pressing power key (or home key) from any modes, the host will reset the IC according to the Reset timing and implement the reset process.

7.4 Recommended to Apply in Conjunction with IR

If gesture wake-up function is applied in conjunction with IR, the host can enable GT1151Q to enter Sleep mode to reduce power consumption when IR detects shielding object while screen-off. Otherwise, GT1151Q enters Gesture mode. To enter different modes, use the methods described above (reset is required before sending command).

8. Reading Coordinate and Verification in Gesture Mode

8.1 Reading Coordinate

In Gesture mode, when 0x814C is not 0, the host can acquire the wakeup trajectory of user by reading the gesture features and gesture coordinates.

The gesture protocol type: the host obtains the gesture protocol type by reading the register 0x814E.

The supported protocol types are shown below:

bit7~bit4(auxiliary information):

0x00: NULL;

0x01: Reserved;

0x02: Breaking points of multi-stroke gesture.

bit3~bit0 (Gesture coordinates):

0x00: NULL;

0x01: Reserved;

0x02: the touch points of the gesture trajectory extracted based on certain spacing.

0x03: feature points of the gesture trajectory(the feature points is extracted according to the gesture type.

For example: the feature points of the “W” trajectory are: starting point, three turning points and end point.)

Gesture auxiliary information: the host obtains the length of the auxiliary information by reading the register 0x814F and then acquires the auxiliary information by reading the registers from 0x8150 to 0x81A0 according to the length.

Multi-stroke gesture breaking points: the end point of each stroke within a multi-stroke gesture. The ordinals of the strokes start from 0.

Gesture coordinates: the host obtains the number of coordinates of the gesture trajectory by reading the register 0x814D and then reads the registers from 0xBDA8 to 0xBEA7 according to the principle that

every four registers correspond to one coordinate. By synthesizing the above information, the host obtains the accurate touch trajectory of user.

8.2 Coordinate and Trajectory Information Verification

To improve the system reliability, data transmitted between GT1151Q and the host should be verified; if the data is read by the host, the data should be read for several times.

There are two kinds of verifications. One is implemented when the extreme points of the gesture are reported; the other is implemented when the complete trajectory coordinates of the gesture are reported. When receiving interrupt that indicates reading gesture information, the host will check whether the MSB at 0x814E is 1. If yes, the two verifications listed below are required. If no, verification is not required and the host can read the coordinate based on the steps described in section 8.1.

a) Gesture extreme point coordinate verification (8-bit verification)

Starting from 0x814C, read consecutive (buffer length+5) u8 data bytes. If the sum of the data bytes is 0, verification succeeds; otherwise, verification fails and the host needs to read the coordinate again. If the verification still fails, the data read this time will be abandoned and 0x814C will be reset to 0 (FW will return to gesture recognition state when 0x814C is 0).

b) Gesture trajectory coordinate verification (16-bit verification)

Starting from 0xBDA8, read consecutive (number of touch points *2) u16 data bytes and then read the 16-bit checksum at 0x819F. If the sum of the data bytes and checksum is 0, verification succeeds; otherwise, verification fails and the host needs to read the trajectory coordinates again. If the verification still fails, the data read this time will be abandoned and 0x814C will be reset to 0 (FW will return to gesture recognition state when 0x814C is 0).

Note: 0x819F stores the low bits of the checksum while 0x81A0 stores high bits. The MSB of 0x814E as 1 indicates coordinate verification is required.

9. Revision History

Revision	Date	Description
Rev.01	2014-07-17	Initial release